HARD · ADVANCED MATH

Equivalent expressions

03

9 questions · ~14 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ADVANCED MATH

In the expression $3(2x^2+px+8)-16x(p+4)$, p is a constant. This expression is equivalent to the expression $6x^2-155x+24$. What is the value of p ?

A. -3

B. 7

C. 13

D. 155

Q3 ADVANCED MATH

$$9x^4 + 8x^3 - 7x^3 + 5x^2 - 3x^2 + x$$

The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a?

A. -8

B. -2

C. 2

D. 8

Q4

ADVANCED MATH

$$\frac{x^2 - c}{x - b}$$

In the expression above, b and c are positive integers. If the expression is equivalent to $x + b$ and $x \neq b$, which of the following could be the value of c ?

A. 4

B. 6

C. 8

D. 10

Q5

ADVANCED MATH

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$?

A. $a + c$

B. $a^2 + c^2$

C. $a^2 + 2ac + c^2$

D. a^2c^2

Q6

ADVANCED MATH

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 28x + 14b$

C. $3x^2 + 42x + 14b$

D. $3x^2 + 49x + 14b$

Q7

GRID-IN ADVANCED MATH

The expression $3x - 2319x + 6$ is equivalent to the expression $ax^2 + bx + c$, where a, b, and c are constants. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ADVANCED MATH

If $a = c + d$, which of the following is equivalent to the expression $x^2 - c^2 - 2cd - d^2$?

A. $(x + a)^2$

B. $(x - a)^2$

C. $(x + a)(x - a)$

D. $x^2 - ax - a^2$

Q9 ADVANCED MATH

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $hx + kx + j$, where h, k, and j are integer constants. Which of the following must be an integer?

A. $\frac{b}{h}$

B. $\frac{b}{k}$

C. $\frac{45}{h}$

D. $\frac{45}{k}$